



Information Technology Institute (ITI)
Analog IC Design Track
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Lab 04
Common Drain Frequency Response

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Part 1: Sizing Chart and Transistor Scaling

1) Introduction & Objective

The objective of this part is to design a PMOS transistor operating at a specific sweet-spot that provides an optimal trade-off between gain, area, and speed. In real MOSFETs, the square-law model is inaccurate due to velocity saturation and mobility degradation. Therefore, we rely on a simulation-based sizing chart using a parameter called V^* , defined as $V^* = \frac{2I_D}{g_m}$. The target specifications for our design are $L = 1\mu m$, $V^* = 200mV$, and a total current consumption of $I_D = 10\mu A$.

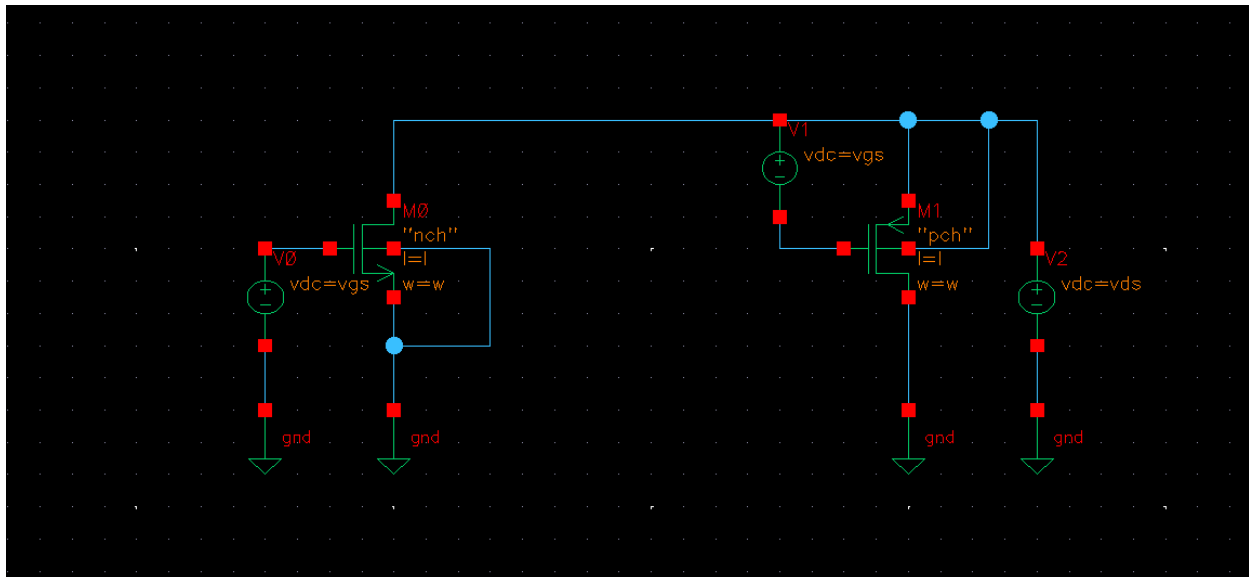


Figure 1:schematic



2) Generating the Sizing Chart

To generate the sizing chart, an initial testbench was created with an arbitrary width $W = 10\mu m$ and the required length $L = 1\mu m$. A DC sweep was performed for V_{GS} from $0V$ to approximately $V_{TH} + 0.4V$. Using the calculator, the expressions for V^* and V_{ov} ($V_{GS} - V_{TH}$) were overlaid on the same plot.

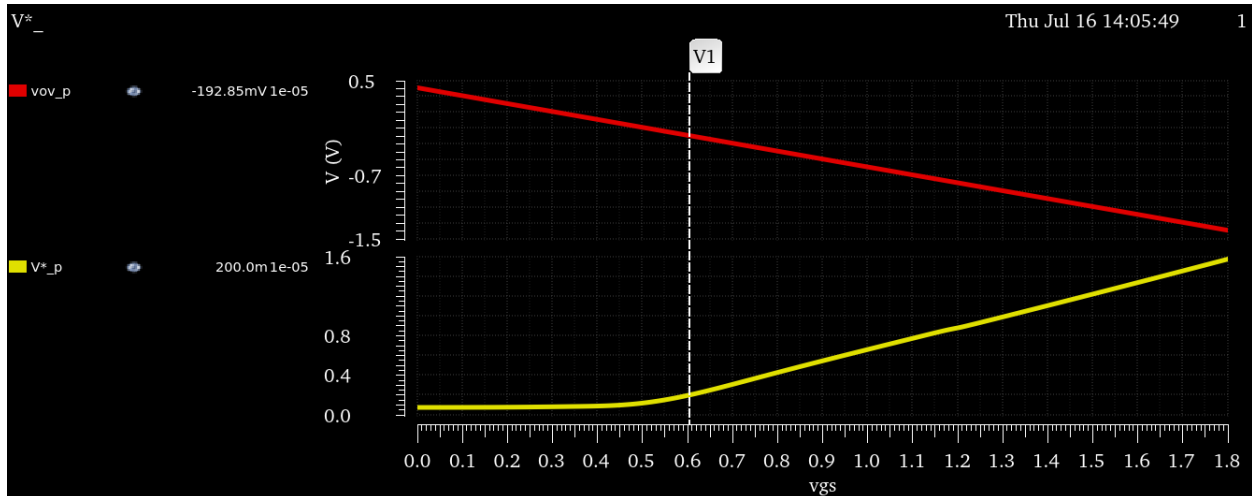


Figure 2: V^* & V_{ov} vs V_{gs}

3) Extracting Operating Point Parameters

($Q - Point$) Based on the design constraints, the target V^* is $200mV$. By locating $V^* = 200mV$ on the generated chart, the corresponding gate-source voltage was found to be $V_{GSQ} \approx 0.6V$. At this specific V_{GSQ} , the values for I_D , g_m , and g_{ds} were extracted from the simulation for the initial $10\mu m$ width.

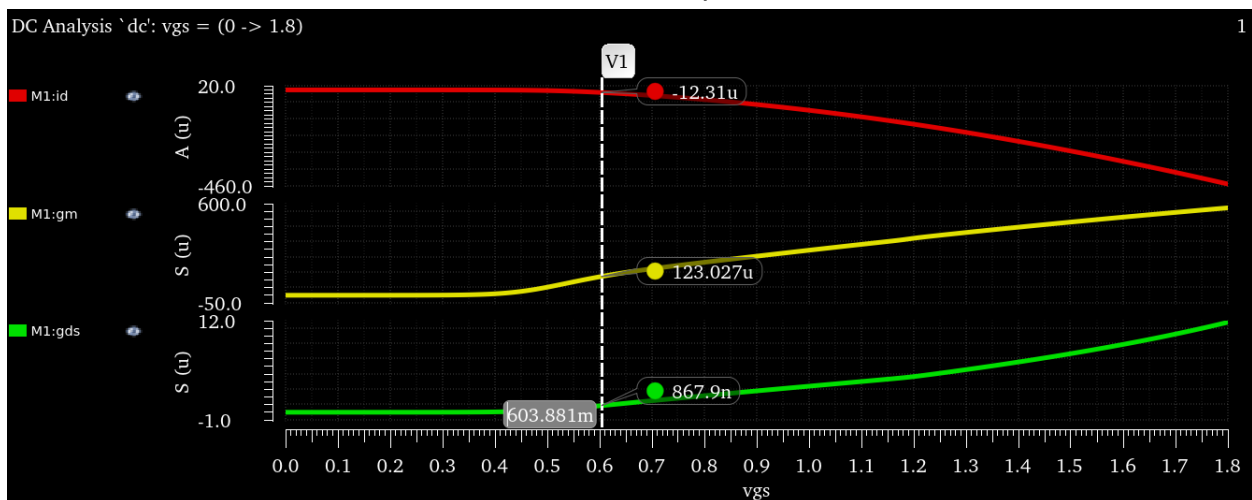


Figure 3: i_d , g_m & g_{ds} vs V_{gs}



Idx	gm	gds
12.3uA	123.027 uS	867.9 nS

4) Transistor Sizing using Ratio and Proportion

Since the drain current (I_D) is directly proportional to the transistor width (W) regardless of the operating region, we can use logical cross-multiplication to determine the exact width required to achieve the target $10\mu A$ current.

$$W_{new} = \frac{I_{DQ} \times W_{initial}}{I_{DX}} = \frac{10\mu A \times 10\mu m}{12.31\mu A} \approx 8.12\mu m$$

5) Final Verification

To verify our logical derivations, the PMOS transistor width was updated to $W = 8.1234\mu m$ and a final DC operating point simulation was executed.

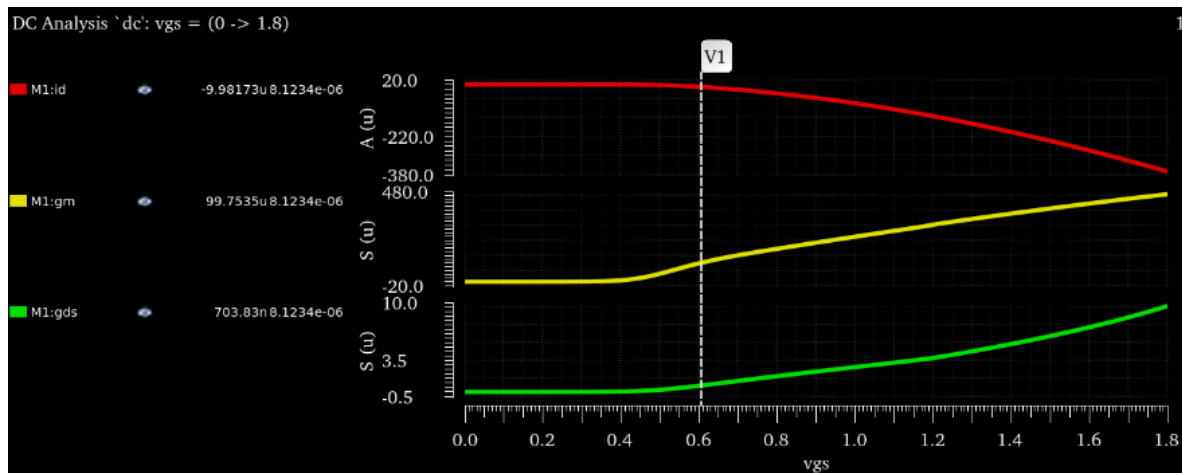


Figure 4: id, gm & gds vs Vgs when $W=8.12\mu m$

The final simulation results perfectly align with the target specifications:

- Target Current Reached: $I_D = 9.98\mu A \approx 10\mu A$
- Transconductance: $g_m = 99.75\mu S$
- Output Conductance: $g_{ds} = 703.83nS$
- Intrinsic Gain: $|A_v| = \frac{g_m}{g_{ds}} = \frac{99.75\mu S}{703.83nS} \approx 141.73$



Part 2: PMOS Common Drain (Source Follower) Amplifier Analysis

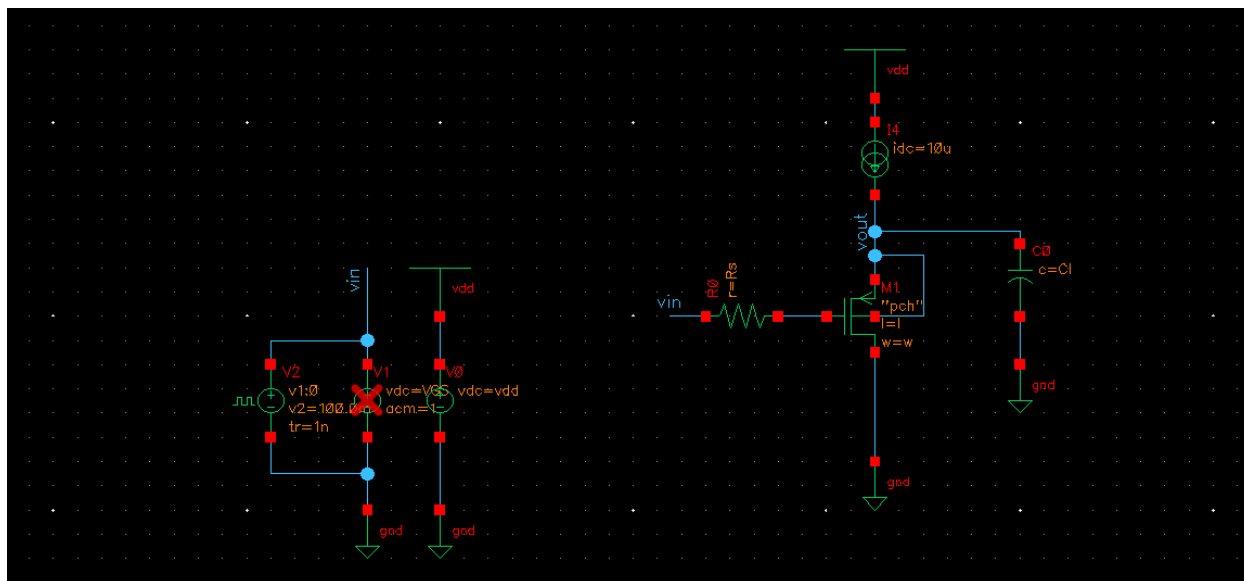


Figure 5: commen_drain

1. DC Analysis & Operating Point Verification

- The objective of this stage is to verify that the transistor is operating in the saturation region and to establish the baseline biasing values required for subsequent high-frequency analysis.

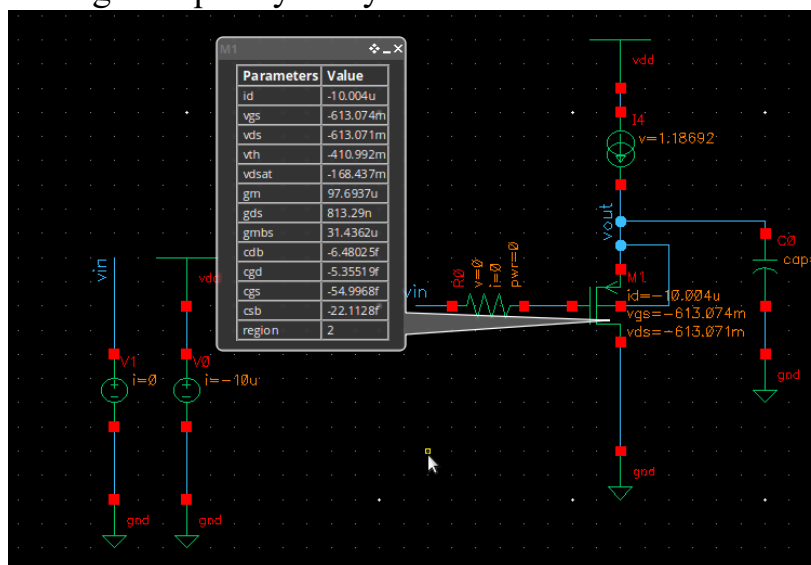


Figure 6:DC_OP



Logical Conclusion:

The transistor is correctly biased in the saturation region (Region 2) using a constant $10\ \mu\text{A}$ current source, as explicitly confirmed by the annotated schematic operating point parameters. This topology ensures a robust and stable transconductance (g_m) that we can rely on for the subsequent high-frequency AC analysis.

2. AC Analysis & Frequency Response

This section investigates the impact of load capacitance (C_L) and source resistance (R_S) variations on the circuit's frequency stability and the emergence of peaking (indicating an underdamped system).

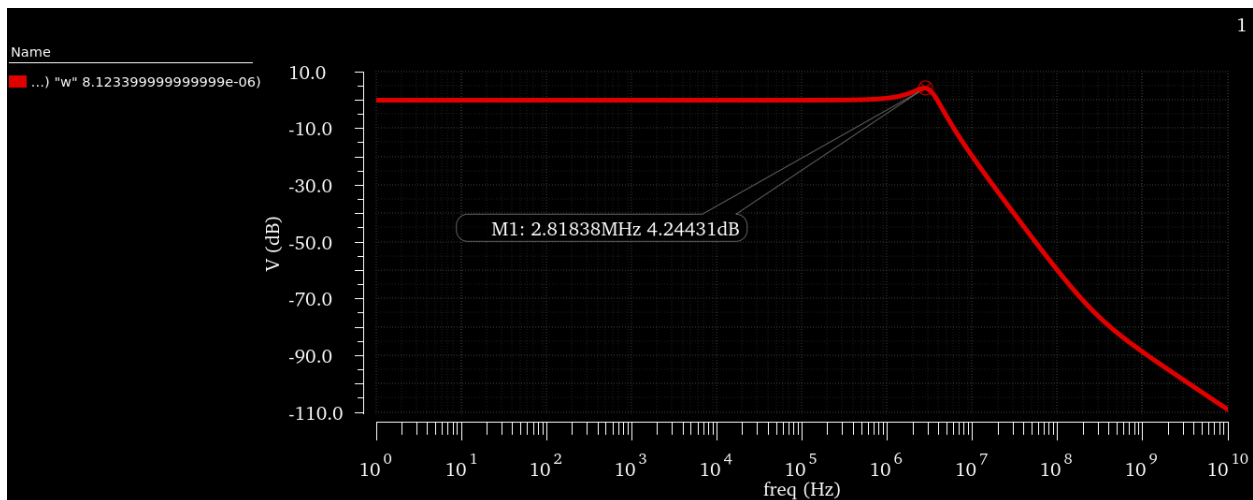


Figure 7:bode_plot

Do you notice frequency domain peaking?

→ Yes, then $Q > 0.707$.

Analytically calculate quality factor (use approximate expressions). Is the system underdamped or overdamped?

→ Special case : R_S high (ideal current source) & ignore CLM & body effect + C_L high.



$$\omega_o = \sqrt{\frac{gm}{(cgs + cgd) \times Cl \times R_{sig}}} \quad \& \quad Q = \sqrt{\frac{gm(cgs + cgd)R_{sig}}{Cl}}$$

$$Q = \sqrt{\frac{97.6937 * 10^{-6} * (55 + 5.35) * 10^{-15} * 2 * 10^6}{2 * 10^{-12}}} = 2.428 > 0.707$$

A. Effect of Load Capacitance (CL) Variation

The source resistance was fixed at $R_S = 2 \text{ M}\Omega$, and a parametric sweep was performed for the load capacitor.

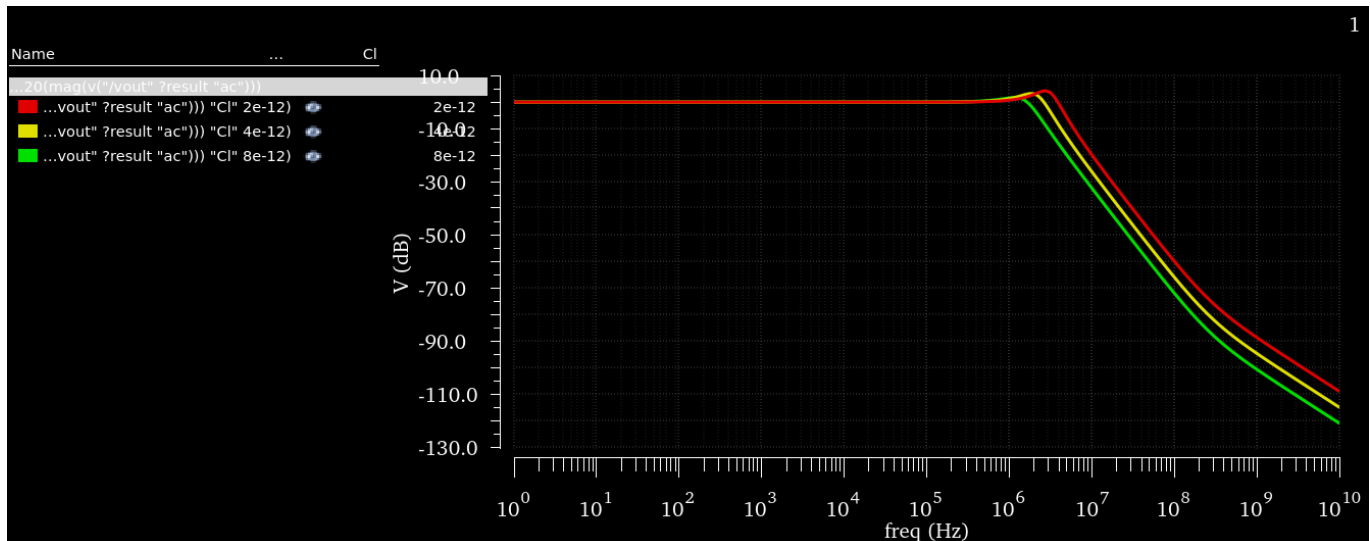


Figure 8: Cl_sweep



Load Capacitance (CL) Peak Magnitude (dB) System State

2 pF	4.244 dB	Highly Underdamped
4 pF	3.144 dB	Underdamped
8 pF	1.640 dB	Less Underdamped

Step-by-Step Analysis: Increasing the load capacitance (CL) physically increases the damping of the system. Mathematically, this heavier damping reduces the Quality Factor (Q), which directly suppresses the peaking magnitude in the frequency response.

B. Effect of Source Resistance (RS) Variation

The load capacitance was fixed at $CL = 2 \text{ pF}$, and a parametric sweep was performed for the source resistance.

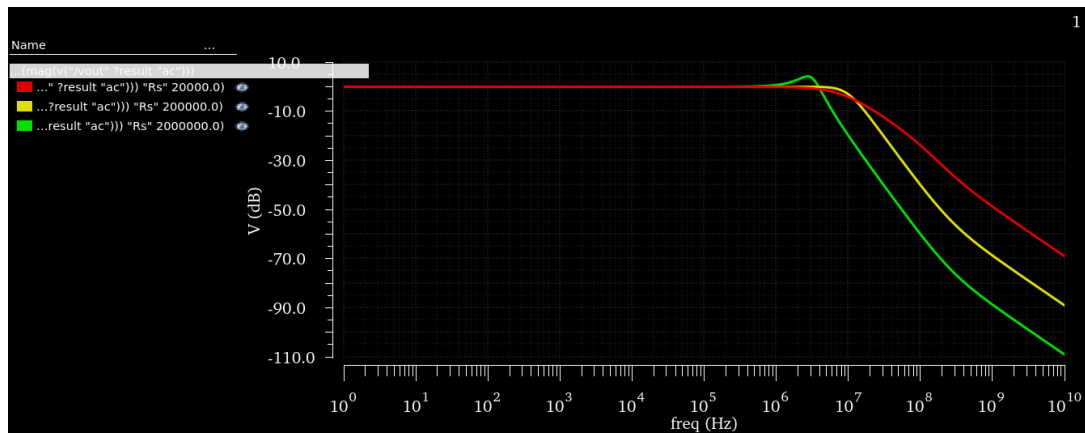


Figure 9:RSig_sweep



Source Resistance	Peak Magnitude (dB)	System State
20 k Ω	-0.072 dB (~ 0 dB)	Overdamped / Critically Damped
200 k Ω	-0.060 dB (~ 0 dB)	Overdamped / Critically Damped
2 M Ω	4.244 dB	Underdamped

Step-by-Step Analysis: A larger source resistance (R_S) interacts more significantly with the intrinsic transistor capacitances (C_{gs} and C_{gd}). This interaction generates complex poles, elevating the system's overall Q-factor and causing pronounced peaking in the Bode plot.

3. Transient Analysis (Time Domain)

To validate the AC stability conclusions, a step input was applied to observe the circuit's physical response in the time domain.

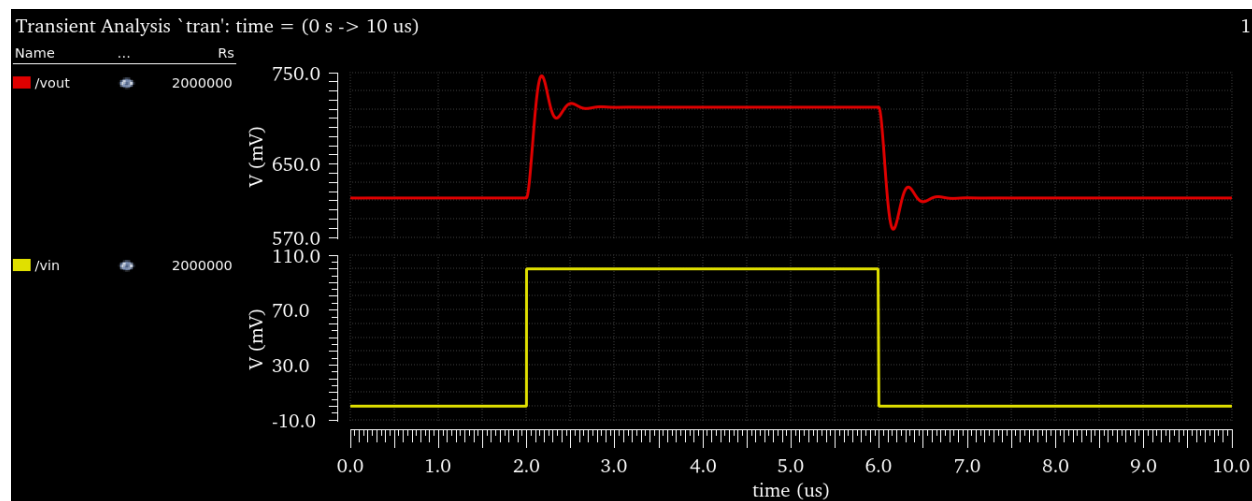


Figure 10: trans_response



1) DC Voltage Difference (DC Shift) Between V_{in} and V_{out}

Calculated DC Shift & Relation to V_{GS} :

In a Source Follower topology, the output is taken at the source terminal, and the input is applied at the gate terminal. The DC shift is mathematically defined as $V_{out(DC)} - V_{in(DC)}$. Since $V_{out} = V_S$ and $V_{in} = V_G$, the shift is exactly equal to $V_S - V_G = -V_{GS}$.

For our PMOS device operating in saturation, V_{GS} is inherently negative. Consequently, $-V_{GS}$ yields a positive value, meaning the signal is shifted UP by an amount equal to $|V_{GS}|$.

How to shift the signal down:

To shift the DC level of the signal down instead of up, an NMOS Source Follower must be utilized. In an NMOS device, the gate voltage must be physically higher than the source voltage to maintain the saturation region ($V_{GS} > 0$). This characteristic dictates that $V_{out} = V_{in} - V_{GS}$, effectively pulling the output DC level down relative to the input.

2) Time Domain Ringing and Overshoot

Observation:

Yes, time-domain ringing is visibly noticed when the system operates in an underdamped state (specifically at high R_{sig} and low C_L values). Logically, this ringing occurs due to the emergence of complex conjugate poles in the system's transfer function, causing an oscillating energy exchange before the output settles.

Overshoot Calculation:

overshoot(VT("/vout") 1u t 5u t t "time") x	
time (s)	oversho...000000
1 2.000E-6	34.73

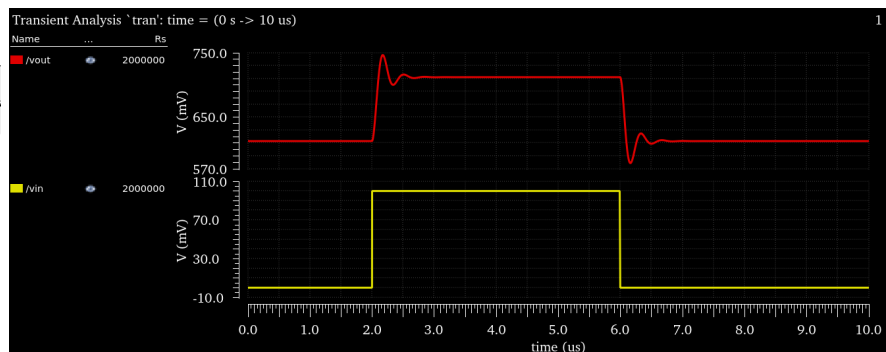


Figure 11: transient analysis input & output



3) Parametric Sweep: Load Capacitance ($C_L = 2p, 4p, 8p$)

Vout vs Time Overlaid Plot:

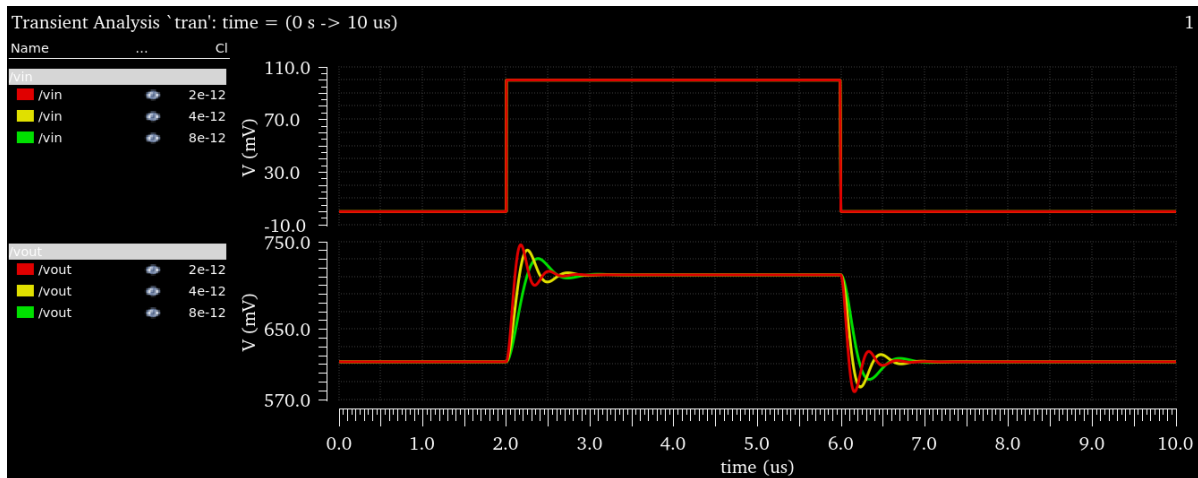


Figure 12: transient C_L sweep

Overshoot vs C_L Plot:

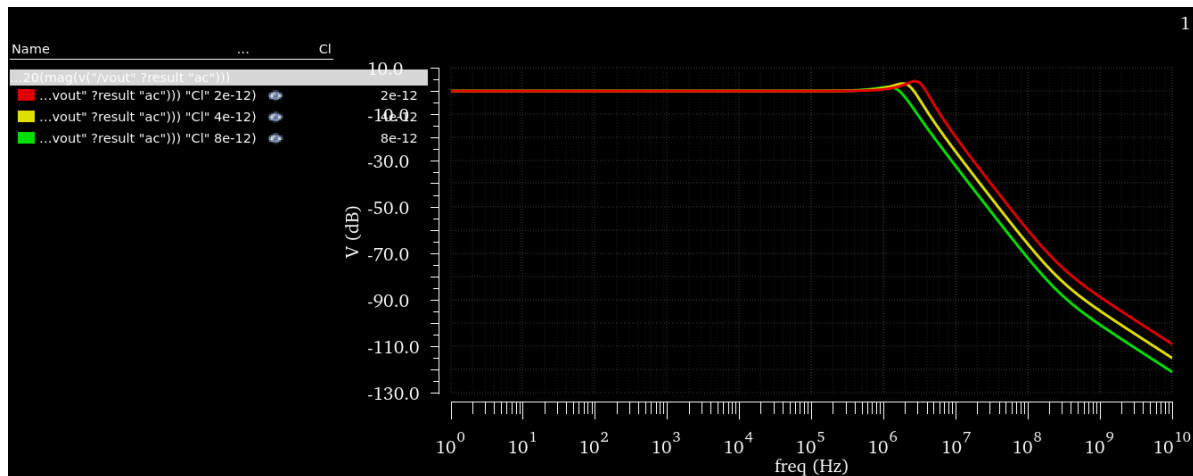


Figure 13: peaking_ C_L sweep

	C_L	$y_{\max}(\text{dB}20(\text{mag}(\text{V}(\text{"/vout"} ? \text{result "ac"}))))$
1	2.000E-12	4.244
2	4.000E-12	3.144
3	8.000E-12	1.640



Comment: As the load capacitance (C_L) increases, the maximum overshoot decreases. From a first-principles perspective, a larger load capacitance demands more charge to change its voltage, which inherently slows down the system and adds physical damping. This heavier damping lowers the overall Quality Factor (Q) of the circuit, forcing the step response to transition from a highly underdamped state (high ringing) towards a more stable, critically damped behavior.

6) Parametric Sweep: Source Resistance ($R_{sig} = 20k, 200k, 2M$)

Vout vs Time Overlaid Plot:

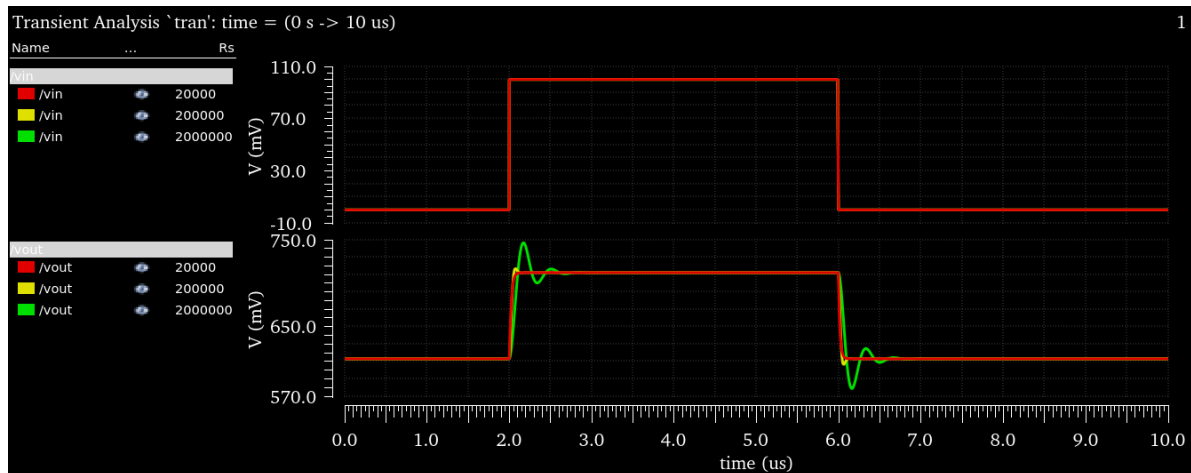


Figure 14: vout vs time $_Rs_sweep$

Overshoot vs R_s Plot:

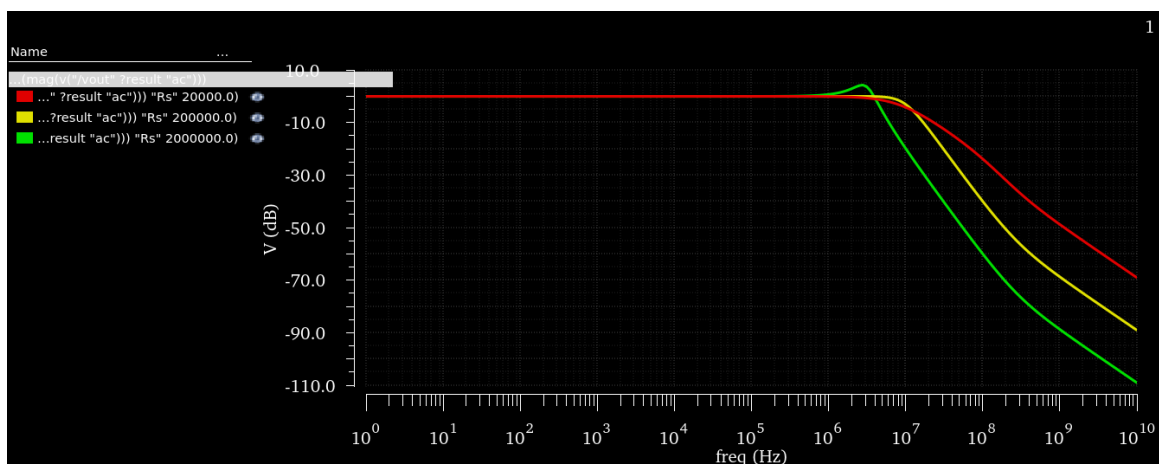


Figure 15: mag. response $_Rs_sweep$



Comment:

- **Peaking and Stability:** The Bode plot explicitly confirms the time-domain observations. The green curve ($R_{sig} = 2M\Omega$) exhibits a distinct magnitude peak near the cutoff frequency. From a circuit design perspective, this large source resistance interacts heavily with the intrinsic gate-to-source capacitance (C_{gs}), shifting the system poles into complex conjugate locations (an underdamped system). In contrast, the red ($20k\Omega$) and yellow ($200k\Omega$) curves show a flat response with no peaking, indicating an overdamped or critically damped system where the poles remain real.
- **Bandwidth Trade-off:** As R_{sig} increases, the -3dB cutoff frequency visibly shifts to the left. This demonstrates a fundamental analog design trade-off: driving the Source Follower with a higher source impedance not only degrades stability (introducing peaking) but also significantly reduces the operational bandwidth of the amplifier.

4) Z_{out} inductive rise

1) Plot the output impedance (magnitude and phase) vs frequency. Do you notice an inductive rise? Why?

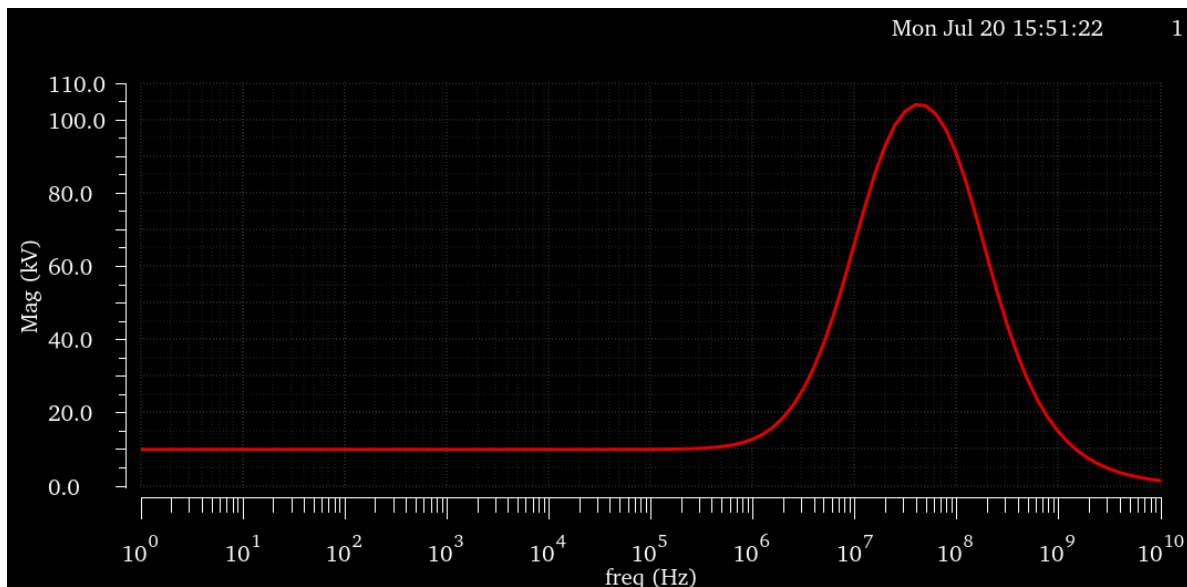


Figure 16: Z_{out} mag.

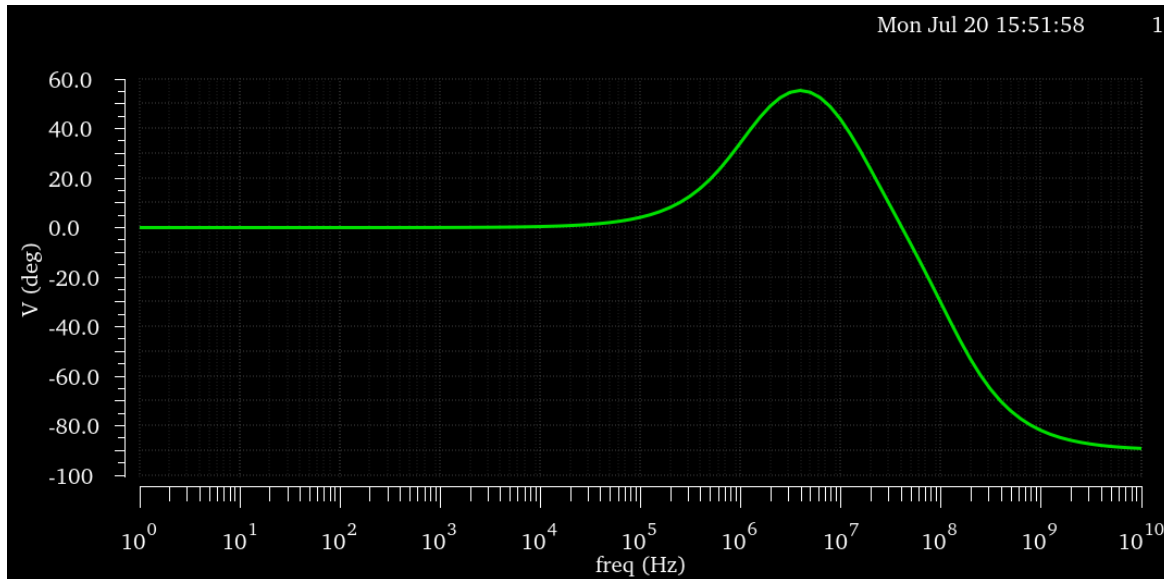


Figure 17: Z_{out} phase

Observation: The simulation shows an inductive rise in the output impedance (Z_{out}) magnitude starting around 1 MHz, peaking near 40 MHz at approximately 105 k Ω . The phase also shifts from 0° towards a positive peak of +55°, confirming the inductive nature.

Why it happens: This inductive behavior is caused by the interaction between the source resistance (R_{sig}) and the gate-source capacitance (C_{gs}). Looking into the source terminal, the impedance appears as a resistance in series with an equivalent inductance $L_{eq} \approx \frac{C_{gs} \cdot R_{sig}}{g_m}$. As frequency increases, the impedance of this equivalent inductor increases, causing the observed rise in Z_{out} .

2) Does Z_{out} fall at high frequency? Why?

Beyond 40 MHz, Z_{out} falls rapidly, and the phase approaches -90° , showing purely capacitive behavior.

Why it happens: At high frequencies, two effects dominate:



1. The gate-drain capacitance (C_{gd}) appears in parallel with R_{sig} . As frequency increases, C_{gd} acts as an AC short, effectively bypassing R_{sig} to ground and destroying the inductive effect.
2. The output node is shunted to ground by the total output capacitance (primarily the source-bulk capacitance C_{sb} and the bypassed C_{gs}). This forms a low-impedance path to ground, making $Z_{out} \approx \frac{1}{s(C_{gs} + C_{sb})}$, which approaches zero at infinite frequencies.

3) Analytical vs. Simulation Comparison

Based on the small-signal model, the critical zeros, poles, and asymptotes of Z_{out} are analytically calculated and compared to the Virtuoso AC simulation results in the following table:

Parameter	Analytical Expression	Simulation Result (From Plot)
Low-Freq Magnitude	$Z_{out} \approx \frac{1}{g_m} = 10.236k\Omega$	$\approx 10\text{ k}\Omega$
Inductive Zero (f_z)	$f_z \approx \frac{1}{2\pi R_{sig}(C_{gs} + C_{gd})} = 1.31MHz$	Magnitude starts rising $\approx 1\text{ MHz}$
Dominant Pole (f_p)	$f_p \approx \frac{g_m}{2\pi(C_{gs} + C_{sb})} = 201\text{ MHz}$	Peak magnitude occurs at $\approx 200\text{ MHz}$
High-Freq Behavior	$Z_{out} \approx \frac{1}{s(C_{gs} + C_{sb})}$	Drops asymptotically towards 0Ω